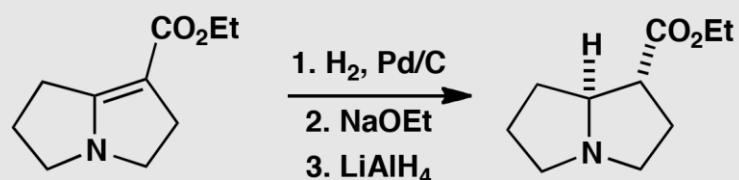


环状化合物非对应选择性+不对称合成

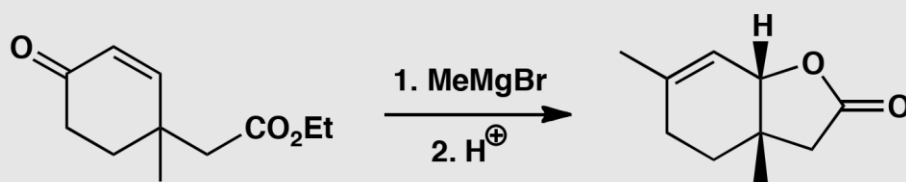
1. Note: 产物改为醇

Explain the stereochemistry of this sequence of reactions, noting the second step in particular.



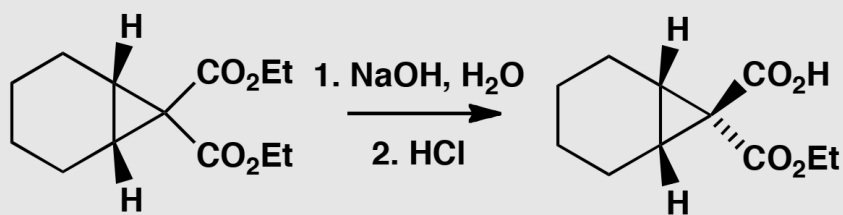
2.

What controls the stereochemistry of this product? You are advised to draw the mechanism first and then consider the stereochemistry.



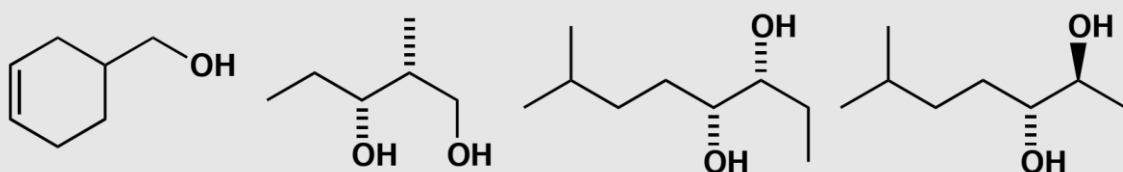
3.

Why is one of these esters more reactive than the other?



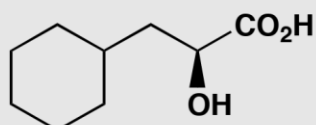
4.

How would you make enantiomerically enriched samples of these compounds (either enantiomer)?



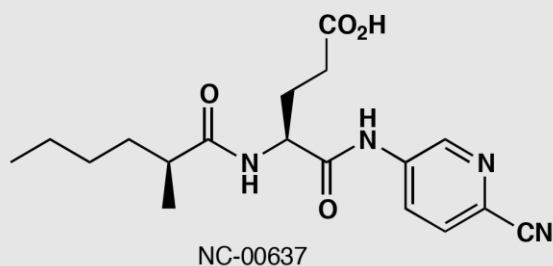
5.

This compound is a precursor to a Novartis drug used for the control of inflammation. How might it be made from a chiral pool starting material?



6.

This compound was developed by the Nutrasweet company as an artificial sweetener. Propose a strategy for its synthesis. Would your proposed approach still be suitable if the compound had turned out to be a successful product, required in multi-tonne quantities?



7.

The two aldehydes below are valuable products in the perfumery industry (Tropional® is a component of Issey Miyake's *L'Eau d'Issey* and Florhydral® is a component of *Allure* by Chanel). How would you make them as single enantiomers?

